

CANVAS EXPORT

# System flow assessment

Product completeness of DECSI Loan Hub against the 12-stage diagram in presentation/system\_flow.png. Code and automated-test review — not a live production UAT at DECSI.

**91.2% flow completeness**

Mean of 12 stage scores (0–100). 1,094 / 1,200 completeness points (was 90.5% / 1,086 on 20 Aug).

**9 / 12 fully implemented**

Application through appraisal, authorization, workflow, and analytics.

**1 config · 2 partial**

Disbursement mock until CBS. Legal admin and dedicated fraud still partial.

**276 / 276 tests**

Clean gate on 21 Aug 2026 re-run. docker compose exec web python manage.py test

SOURCE CANVAS

**system-flow-assessment.canvas.tsx**

DATE

**21 August 2026**

DIAGRAM

**presentation/system\_flow.png**

CLASSIFICATION

**Seqela internal — do not send as-is**

1. Overall

**The credit factory is built.** Weighted completeness is 91.2% across the 12 flowchart boxes (was 90.5% on 20 Aug). Nine stages are fully implemented in product code. Disbursement to core banking is coded, with a Finance CBS payload preview, but still mock-by-default until DECSI\_BASE\_URL is set. Two boxes are only partial: standalone legal administration, and a dedicated fraud-detection product.



Coverage of the 12-box flow

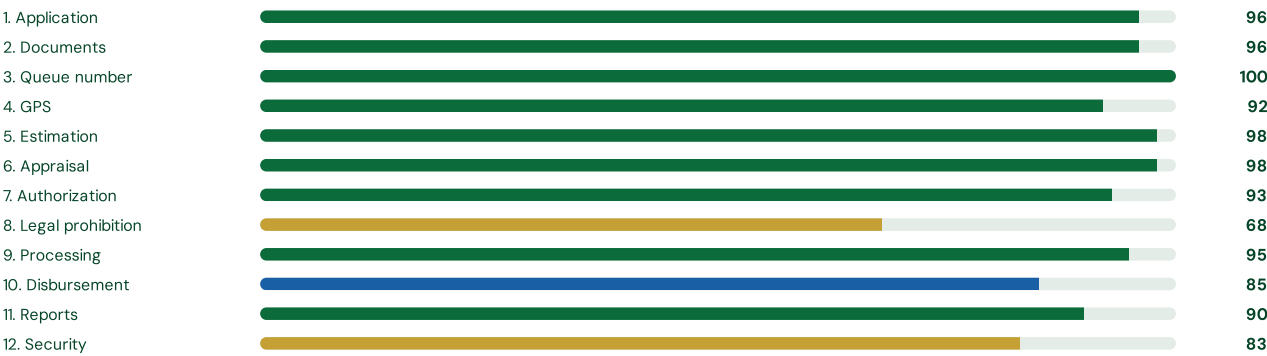


Stage verdict mix



Count of flowchart boxes · n = 12 · Source: code review 21 Aug 2026.

Completeness by stage (%)



Score 0–100 per flowchart box · mean 91.2%. Gold = partial · blue = config. Re-assessed 21 Aug 2026.

## 2. Stage-by-stage result

Scores are reviewer estimates of flowchart intent vs shipped code, not production KPIs.

| # | FLOWCHART BOX                                                | VERDICT           | SCORE | WHAT THE PRODUCT DOES                                                                                                                             | EVIDENCE                                                                                       | GAP                                                                   |
|---|--------------------------------------------------------------|-------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1 | Application<br>Applying for loan; getting information        | FULLY IMPLEMENTED | 96    | Staff create plus Digital Apply. Mock vs live source labelled; silent mock refused when live CBS URL is set; Chapa callback gated to public HTTPS | profile_data_source;<br>callback_url_is_public_https;<br>email_health                          | CBS customer lookup still uses demo mock when DECSI_BASE_URL is unset |
| 2 | Documents<br>Necessary documents; get approval               | FULLY IMPLEMENTED | 96    | Per-loan-type checklist, Eng+Amh OCR, integrity/identity checks, officer verify/reject, default extraction maps that fill appraisal Sheet 1       | document_auth.py;<br>document_extraction_defaults.py;<br>sync_sheet1_from_documents            | LLM and external ID checks are optional hooks, not always on          |
| 3 | Loan Queue<br>No Generation                                  | FULLY IMPLEMENTED | 100   | HK-##### issued on staff, online, agent, and CSV create                                                                                           | generate_incremental_loan_request_id();<br>LatestLoanRequestID                                 | No separate ticket-printer app — the queue number is loan_request_id  |
| 4 | GPS<br>Collateral integration; data capture                  | FULLY IMPLEMENTED | 92    | Field visits for building/land/other: site GPS, photo EXIF, maps, offline tablet sync, evidence pack                                              | collateral/views field_visit;<br>site_gps_lat/lon; /collateral/loan/<id>/...                   | Gebeta tiles need an API key; OSM fallback works without it           |
| 5 | Collateral Estimation<br>Estimation detail; estimated amount | FULLY IMPLEMENTED | 98    | BOQ catalog, unit prices, building/land/movable valuation, engineering QA, grand total into appraisal Sheet 5                                     | BuildingValuation, LandValuation,<br>compute_collateral_totals;<br>/collateral/engineering-qa/ | None material                                                         |
| 6 | Loan Appraisal<br>Financial, business, approval              | FULLY IMPLEMENTED | 98    | 7-step MSME and corporate wizard. Sheet 1 labelled Mock vs Live and prefilled from documents                                                      | APPRAISAL_STEPS;<br>apply_document_extractions;<br>loan_appraisal_step1.html                   | Sheet 7 (repayment schedule) is optional by design                    |

| #  | FLOWCHART BOX                                                   | VERDICT                         | SCORE | WHAT THE PRODUCT DOES                                                                                                                                               | EVIDENCE                                                                          | GAP                                                                                                  |
|----|-----------------------------------------------------------------|---------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 7  | Authorization Branch / zonal / HO operation / HO mgt            | FULLY IMPLEMENTED               | 93    | Configurable committees: branch → district → head_office → management, amount bands, votes, delegations                                                             | ApprovalCommitteeLevel keys; loans/committee.py; /hub/manage_approval_committees/ | Flowchart "Zonal" is District; "HO Operation" is head_office (Cooperative intake is a separate gate) |
| 8  | Prohibition of collateral in legal administration               | PARTIALLY IMPLEMENTED           | 68    | Post-approval legal papers that block disbursement: restriction, POA, title search, mortgage, notary                                                                | loans/collateral_legal.py; /hub/loan_request/<id>/post_approval/                  | No Legal role, legal queue, or legal_cleared stage — enforced as a finance gate, not its own module  |
| 9  | Processing Workflow Status tracking; integration & verification | FULLY IMPLEMENTED               | 95    | Role queues, committee status, document verify, process policy, risk desk, notifications, remote SMS OTP agreement signing, delegation document access              | remote_sign.py; /sign/agreement/<token>; StaffDelegation                          | Workflow is Django state machines, not a BPMN engine                                                 |
| 10 | Disbursement Disbursing to core banking                         | CODE COMPLETE, LIVE CBS PENDING | 85    | Post-approval → Finance → confirm → book. Finance sees CBS payload preview and a customer-number gate. Live Temenos client exists; mock ledger is still the default | finance_ready_to_book; cbs_client.build_disbursement_payload                      | Live booking needs DECSI_BASE_URL; DECSI_CBS_USE MOCK_LEDGER defaults to True                        |
| 11 | Report & Analytics                                              | FULLY IMPLEMENTED               | 90    | Classic reports plus Credit Intelligence (pipeline, collateral, decision, assistant) and book-ops desks                                                             | /hub/credit-intelligence; reporting.py; views_book.py                             | NPL / outstanding KPIs stay adapter-backed and weak until live CBS                                   |

| #  | FLOWCHART BOX                                                | VERDICT               | SCORE | WHAT THE PRODUCT DOES                                                                                                                                                               | EVIDENCE                                                                         | GAP                                                                                            |
|----|--------------------------------------------------------------|-----------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 12 | Security and Compliance RBAC; authorization; fraud detection | PARTIALLY IMPLEMENTED | 83    | 20+ roles, MFA, lockout, audit log, committee bands, password min-length 10, nosniff/DENY headers, document "suspicious" verdicts, watchlist, risk desk, delegation document access | CustomUser.ROLE_CHOICES; EnforceMfaMiddleware; SecurityAuditLog; StaffDelegation | No dedicated fraud/AML case engine or transaction-fraud model — document auth + watchlist only |

### 3. Flowchart labels vs product names

Authorization titles in the diagram do not match the committee keys in code. Functionally the four-level chain is there.

| SYSTEM_FLOW.PNG                                   | PRODUCT IMPLEMENTATION                                              | MATCH     |
|---------------------------------------------------|---------------------------------------------------------------------|-----------|
| Branch (S/branch) Level                           | ApprovalCommitteeLevel key = branch (sub-branch via Branch)         | Direct    |
| Zonal Level                                       | key = district (District Manager / district committee)              | Renamed   |
| HO Operation Level                                | key = head_office committee; Cooperative intake is a separate queue | Split     |
| HO Mgt Level                                      | key = management (CEO / VP / Board)                                 | Direct    |
| Prohibition of collateral in legal administration | Post-approval legal documents + disbursement blockers               | Relocated |
| Disbursing to Core Banking system                 | cbs_client + Finance gate; mock ledger unless DECSI_BASE_URL        | Config    |
| Fraud Detection                                   | Document OCR suspicious verdict + watchlist + risk desk             | Partial   |

## 4. Gaps that keep the flow from 100%

**8. Legal administration is a finance gate, not a module · 68%**

The diagram places prohibition of collateral after authorization as its own step. The product stores restriction, power of attorney, title search, mortgage registration, and notary papers on the post-approval screen and blocks mark-disbursed until they are verified. There is no lawyer role, legal queue, or legal\_cleared status. Closing the gap means a Legal Administration queue between committee\_approved and Finance.

**12. Fraud detection is document-level, not a fraud product · 83%**

Role-based access, committee authorization bands, TOTP MFA, session lockout, password minimum length 10, nosniff / X-Frame-Options DENY, and SecurityAuditLog are implemented. What the flowchart calls fraud detection is OCR integrity, identity match, a suspicious document verdict, a loan watchlist, and the risk desk. There is no AML case system, device/velocity engine, or transaction-fraud model.

**10. Core banking disbursement is mock until DECSI\_BASE\_URL · 85%**

The path post-approval → Finance approval → officer confirm → CBS book is fully coded (finance\_ready\_to\_book, cbs\_client.py, tests). Finance now sees a CBS payload preview and is blocked without a customer number. Default settings use DECSI\_CBS\_USE MOCK\_LEDGER=True. Live Temenos booking requires DECSI\_BASE\_URL. Customer lookup refuses silent mock when a live URL is set.

**Adjacent: Seqela Market staff admin is unwired · portal only**

The external market portal at /market-portal/ is live. Staff market admin views exist under partners/ but are not mounted in the root URL conf (tests expect /partners/ to 404). This is outside the 12 boxes but is part of the surrounding product.

## 5. Automated test run

**Clean gate: 276 / 276 passed.** Re-run in the web container on 21 Aug 2026 after the feature update (remote OTP, document extraction, finance ready-to-book, Chapa HTTPS gate). One Nominatim geocode warning (network unreachable) did not fail any test.

| 276<br>Tests run                                                              | 276<br>Passed | 0<br>Failed                                                           | 90.9s<br>Duration |
|-------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------|-------------------|
| WHAT CHANGED SINCE 20 AUG                                                     |               | EFFECT ON THIS ASSESSMENT                                             |                   |
| Remote SMS OTP agreement signing (/sign/agreement/<token>/)                   |               | Stage 9 stronger (closing pack); stage 10 still mock CBS              |                   |
| Document extraction defaults fill appraisal Sheet 1; Mock vs Live labels      |               | Stages 1, 2, 6 scored up; live CBS still required for real party data |                   |
| Finance ready-to-book + CBS payload preview; no silent mock when live URL set |               | Stage 10 82 → 85; still config until DECSI_BASE_URL in the target env |                   |
| New tests: test_decsei_short_list, test_document_extraction_and_api_labels    |               | Suite 250 → 276, all passing                                          |                   |

Pass rate 100% (276/276) · docker compose exec web python manage.py test

## 6. Closing the flowchart

1. Add a Legal Administration queue and role so stage 8 sits between committee approval and Finance, matching the diagram.
2. Point DECSI\_BASE\_URL at core banking and turn off mock ledger in the target environment so stage 10 is a live disbursement.
3. If the diagram's "Fraud Detection" is a contractual requirement, specify the product (document-only vs AML/transaction) — current coverage is origination fraud controls only.
4. Automated tests are a clean gate (276/276). Remaining work is process design (legal desk, fraud product) and CBS go-live, not test repair.

**Not scored here:** DECSI production currently runs queue and admin only; this assessment is of the enhanced credit-factory codebase against system\_flow.png.



Source diagram: presentation/system\_flow.png

Export of Cursor canvas system-flow-assessment.canvas.tsx · 21 August 2026 · Seqela internal.  
Client-facing pack (no scores / file paths): DECSI\_System\_Flow\_Assessment.pdf